

Project: ITSMDeskPro

IT Service Management & Help Desk Platform

Domain: Communications, Media & Technology

1. Introduction

ITSMDeskPro is a web-based IT service management and help desk platform designed for IT departments, managed service providers, and technology operations teams to manage incident tickets, service requests, problem investigations, change management, asset tracking, and service level compliance. The system enables end users to raise IT support tickets and track resolution, first and second-line support engineers to manage incident queues and escalations, change advisory board members to review and approve change requests, asset managers to maintain the IT asset inventory, and administrators to configure SLA rules, escalation matrices, and service catalog items.

It uses a REST API-based backend architecture with Angular/React frontend and is compatible with Java (Spring Boot) and .NET (ASP.NET Core) platforms.

Actors / Users

- **End User:** Raises IT incident or service requests, tracks ticket status, and provides satisfaction feedback
- **L1 Support Engineer:** Handles first-line triage, resolves standard incidents, and escalates complex issues
- **L2 / L3 Support Engineer:** Investigates escalated incidents, performs root cause analysis, and manages problem records
- **Change Manager:** Reviews change requests, chairs CAB sessions, and approves implementation plans
- **IT Asset Manager:** Maintains the CMDB, tracks hardware and software assets, and manages license compliance
- **ITSM Admin:** Configures service catalog, SLA policies, escalation rules, categories, and user roles

2. Module Overview

- **2.1 Identity & Access Management:** Authentication, RBAC, and audit trails.
- **2.2 Service Catalog & Request Management:** Manages the IT service catalog and handles standard service request fulfillment.
- **2.3 Incident Management:** Tracks IT incidents from logging to resolution with SLA monitoring.
- **2.4 Problem Management:** Manages root cause investigations, known error records, and permanent fixes.
- **2.5 Change Management:** Handles change requests, CAB approvals, and implementation tracking.
- **2.6 IT Asset & CMDB Management:** Maintains IT asset inventory, configuration items, and software license records.
- **2.7 ITSM Analytics & Reporting:** Provides dashboards for ticket volumes, SLA compliance, MTTR, and asset health.
- **2.8 Notifications & Alerts:** Ticket assignment notifications, SLA breach warnings, change approval reminders, and asset expiry alerts.

3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST API-based architecture
- **Database:** Relational Database (MySQL / PostgreSQL / SQL Server)

4. Module-Wise Design

4.1 Identity & Access Management Module

Features

- End user and IT staff registration with role-based access control and team-scoped visibility
- Full audit logging for all ticket actions and configuration changes

Entities

- **User**
 - UserID
 - Name
 - Role
(EndUser/L1Support/L2Support/L3Support/ChangeManager/AssetManager/Admin)
 - Email
 - Phone
 - TeamID
 - LocationID
 - Status (Active/Inactive/Suspended)
- **AuditLog**
 - AuditID
 - UserID
 - Action
 - EntityType
 - RecordID
 - Timestamp

4.2 Service Catalog & Request Management Module

Features

- Maintain a structured IT service catalog with fulfilment workflows per service item
- Process standard service requests with configurable approval and delivery steps

Entities

- **ServiceCatalogItem**
 - ItemID
 - ServiceName
 - Category (Hardware/Software/Access/Network/Other)

- Description
- FulfilmentSLAHours
- ApprovalRequired
- DeliveryTeamID
- Status (Active/Inactive)
- **ServiceRequest**
 - RequestID
 - RequesterID
 - CatalogItemID
 - Details
 - SubmissionDate
 - AssignedToID
 - FulfilmentDueDate
 - Status (Submitted/InProgress/PendingApproval/Fulfilled/Rejected/Cancelled)

4.3 Incident Management Module

Features

- Log, categorise, and assign IT incidents with priority-based SLA enforcement
- Manage escalation tiers, resolution workflows, and user satisfaction scoring

Entities

- **Incident**
 - IncidentID
 - ReporterID
 - Category (Hardware/Software/Network/Access/Application/Other)
 - Description
 - Priority (P1/P2/P3/P4)

- AssignedTeamID
- AssignedToID
- LoggedDate
- ResolutionDate
- SLADueDate
- Status (Open/InProgress/Pending/Resolved/Closed/Reopened)
- **IncidentNote**
 - NoteID
 - IncidentID
 - AuthorID
 - NoteText
 - NoteType (WorkNote/PublicUpdate/Escalation)
 - CreatedDate
- **SatisfactionScore**
 - ScoreID
 - IncidentID
 - UserID
 - Rating (1to5)
 - Feedback
 - SubmittedDate

4.4 Problem Management Module

Features

- Raise problem records linked to recurring or major incidents for root cause investigation
- Maintain known error database and track permanent fix implementation

Entities

- **ProblemRecord**
 - ProblemID
 - LinkedIncidentIDs
 - Title
 - Description
 - AssignedToID
 - Priority (High/Medium/Low)
 - RootCause
 - RaisedDate
 - ResolvedDate
 - Status (Open/UnderInvestigation/KnownError/Resolved/Closed)
- **KnownError**
 - KEID
 - ProblemID
 - Description
 - Workaround
 - PermanentFixETA
 - Status (Active/FixApplied/Closed)

4.5 Change Management Module

Features

- Submit change requests with risk assessment, rollback plans, and implementation windows
- Manage CAB review workflows, approval gates, and post-implementation review

Entities

- **ChangeRequest**
 - ChangeID

- RequestedByID
- ChangeType (Standard/Normal/Emergency)
- Title
- Description
- ImpactAssessment
- RiskLevel (Low/Medium/High/Critical)
- RollbackPlan
- PlannedStartDate
- PlannedEndDate
- Status
(Draft/Submitted/CABReview/Approved/Scheduled/Implemented/PIRPending/Closed/Rejected)
- **CABReview**
 - ReviewID
 - ChangeID
 - ReviewDate
 - AttendeeIDs
 - Decision (Approved/Rejected/DeferredPendingMoreInfo)
 - Comments
 - Status (Scheduled/Completed)
- **ChangeImplementation**
 - ImplementationID
 - ChangeID
 - ImplementedByID
 - ActualStartDate
 - ActualEndDate
 - Outcome (Successful/PartiallySuccessful/Failed/RolledBack)

- PIRComments

4.6 IT Asset & CMDB Management Module

Features

- Maintain hardware and software asset inventory with lifecycle status tracking
- Link configuration items to incidents, changes, and service dependencies

Entities

- **AssetRecord**
 - AssetID
 - AssetType (Laptop/Server/NetworkDevice/Printer/MobileDevice)
 - Make
 - Model
 - SerialNumber
 - AssignedToID
 - LocationID
 - PurchaseDate
 - WarrantyExpiry
 - Status (InUse/InStock/UnderRepair/Retired/Disposed)
- **SoftwareLicense**
 - LicenseID
 - SoftwareName
 - Vendor
 - LicenseType (PerSeat/Concurrent/Enterprise)
 - TotalSeats
 - UsedSeats
 - ExpiryDate

- Status (Active/Expired/UnderRenewal)
- **ConfigurationItem**
 - CIID
 - CName
 - CType
 - LinkedAssetID
 - Owner
 - Environment (Production/UAT/Dev)
 - DependsOnCIIDs
 - Status (Active/Inactive/Decommissioned)

4.7 ITSM Analytics & Reporting Module

Features

- Ticket volume, SLA compliance rate, and MTTR dashboards by team, category, and period
- Change success rate, problem recurrence, asset utilisation, and license compliance reports

Entities

- **ITSMReport**
 - ReportID
 - Scope (Team/Category/Priority/Period)
 - Metrics (TicketCount, SLAComplianceRate, MTTR, FirstCallResolutionRate, ChangeSuccessRate, ProblemRecurrenceRate, LicenseCompliancePercent)
 - GeneratedDate

4.8 Notifications & Alerts Module

Features

- Ticket assignment, update, and resolution notifications for end users and engineers
- SLA breach warnings, change approval reminders, warranty expiry, and license compliance alerts

Entities

- **Notification**
 - NotificationID
 - UserID
 - Message
 - Category (Incident/ServiceRequest/Problem/Change/Asset/SLA)
 - Status (Unread/Read/Dismissed)
 - CreatedDate

5. Deployment Strategy

- **Local:** Angular/React frontend, Spring Boot/.NET Core backend, local DB
- **Production:** Cloud or on-premises with secure APIs and stateless services

6. Database Design

Tables:

- **User**(UserID, Name, Role, Email, Phone, TeamID, LocationID, Status)
- **AuditLog**(AuditID, UserID, Action, EntityType, RecordID, Timestamp)
- **ServiceCatalogItem**(ItemID, ServiceName, Category, Description, FulfilmentSLAHours, ApprovalRequired, DeliveryTeamID, Status)
- **ServiceRequest**(RequestID, RequesterID, CatalogItemID, Details, SubmissionDate, AssignedToID, FulfilmentDueDate, Status)
- **Incident**(IncidentID, ReporterID, Category, Description, Priority, AssignedTeamID, AssignedToID, LoggedDate, ResolutionDate, SLADueDate, Status)
- **IncidentNote**(NoteID, IncidentID, AuthorID, NoteText, NoteType, CreatedDate)
- **SatisfactionScore**(ScoreID, IncidentID, UserID, Rating, Feedback, SubmittedDate)

- **ProblemRecord**(ProblemID, LinkedIncidentIDs, Title, Description, AssignedToID, Priority, RootCause, RaisedDate, ResolvedDate, Status)
- **KnownError**(KEID, ProblemID, Description, Workaround, PermanentFixETA, Status)
- **ChangeRequest**(ChangeID, RequestedByID, ChangeType, Title, Description, ImpactAssessment, RiskLevel, RollbackPlan, PlannedStartDate, PlannedEndDate, Status)
- **CABReview**(ReviewID, ChangeID, ReviewDate, AttendeeIDs, Decision, Comments, Status)
- **ChangeImplementation**(ImplementationID, ChangeID, ImplementedByID, ActualStartDate, ActualEndDate, Outcome, PIRComments)
- **AssetRecord**(AssetID, AssetType, Make, Model, SerialNumber, AssignedToID, LocationID, PurchaseDate, WarrantyExpiry, Status)
- **SoftwareLicense**(LicenseID, SoftwareName, Vendor, LicenseType, TotalSeats, UsedSeats, ExpiryDate, Status)
- **ConfigurationItem**(CIID, CIName, CIType, LinkedAssetID, Owner, Environment, DependsOnCIIDs, Status)
- **ITSMReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **End User Portal:** Ticket submission form, service catalog browser, my tickets tracker, satisfaction rating widget
- **L1 Support Console:** Incident queue, ticket assignment panel, knowledge base search, escalation form
- **L2/L3 Workbench:** Escalated incident queue, problem investigation form, known error editor, RCA document
- **Change Manager Dashboard:** Change request queue, CAB meeting scheduler, approval workflow, PIR tracker
- **Asset Management Console:** Asset registry, CI relationship viewer, license compliance dashboard, warranty expiry tracker
- **Operations Dashboard:** SLA compliance heatmap, MTTR trend chart, first call resolution rate, team workload view

- **Admin Console:** Service catalog builder, SLA policy configurator, escalation matrix, team and category management

8. Non-Functional Requirements

- **Performance:** Support 25,000+ concurrent end users and IT staff across enterprise locations globally
- **Security:** Ticket data access scoped by team and role, RBAC, encrypted audit trails for all configuration changes
- **Scalability:** Multi-team, multi-location architecture supporting enterprise-wide and MSP deployments
- **Availability:** 99.95% uptime with incident and SLA monitoring modules critical to IT operations continuity
- **Maintainability:** Configurable SLA policies, service catalog items, escalation rules, and change workflows without code changes
- **Observability:** Real-time SLA dashboards, ticket pipeline metrics, change calendar, and asset compliance monitoring

9. Assumptions & Constraints

- Phase 1 excludes integration with Active Directory for auto-provisioning and SCCM for automated asset discovery
- Asset data is manually entered; agent-based discovery and auto-import integrations are deferred to Phase 2
- All notifications are in-app only; email ticketing and Teams/Slack bot integrations are out of Phase 1 scope
- SLA calculations are based on business-hours rules configured internally; public holiday calendars are static
- Fully implementable using standard Java Spring Boot / ASP.NET Core, Angular/React, and RDBMS tooling